# Lab 1

**Objective**: Meet the tutoring team, login, and discuss hardware and operating systems

**Exercise 1: Introduction**
Your tutor(s) will introduce themselves, and you can take a moment to meet the other students near you.

**Exercise 2: Logging into the Matrix**
While you're here, please make sure you can log into the computer systems in the lab – you're going to need to use them this semester. If you are having difficulty, please ask for help – use DemCall to summon a tutor/demonstrator.

**Exercise 3: What is a CPU?**
Form groups of about 4 and discuss "What is a CPU?". Things to consider include: The "brain" analogy. What are the following concepts and how do they relate to the CPU: clock speed, core, cache, hyperthreading, SIMD. What is a FLOP and how are FLOPS (FLOPs per second) different from clock speed? Your tutor will get the class back together to discuss your findings.

**Exercise 4: What is a computer?**
In your group discuss "What is a computer?". Things to consider include: What do you need to "build" a computer? Internal components vs external peripherals. How does this differ between a gaming PC and a smartphone? Your tutor will get the class back together to discuss your findings.

**Exercise 5: What is an operating system?**
In your group discuss "What is an operating system?". Things to consider include: What should an OS do? In what ways are Windows, macOS, iOS, and Android similar to, and different from, each other? What about other, more obscure or archaic operating systems? Are there computers that do not need an operating system? Your tutor will get the class back together to discuss your findings.

**Exercise 6: The Windows Task Manager**
Ask one person from your group to open the Windows Task Manager on a lab computer:

> Ctrl+Alt+Delete > Task Manager.

In your group investigate and discuss the information in Windows Task Manager. Start with the Processes tab, and the Performance tab. Why does the Task Manager report CPU usage and memory usage? In what ways might the Process List be useful, for example if you are a system administrator or software engineer? Your tutor will get the class back together to discuss your findings.

**Note:** This lab is assessed, as are the majority of the labs. You need to demonstrate your work and understanding to a tutor / demonstrator. If you can't make it to an assessed lab, please let Chris know.